Proof. According to the definition $\mathcal{P}$ is a homomorphic image of $\tilde{\mathcal{P}}$. The statements of Proposition A1 follow directly from Lemma A1. □

Optimal policies of the state-enriched MDP $\tilde{\mathcal{P}}$ can be transformed to optimal policies of the original MDP $\mathcal{P}$ and, vice versa, each optimal policy of the original MDP $\mathcal{P}$ corresponds to at least one optimal policy of the state-enriched MDP $\tilde{\mathcal{P}}$.

A2.2 Return-Equivalent Sequence-Markov Decision Processes (SDPs)

Our goal is to compare Markov decision processes (MDPs) with delayed rewards to decision processes (DPs) without delayed rewards. The DPs without delayed rewards can but need not to be Markov in the rewards. Toward this end, we consider two DPs $\tilde{\mathcal{P}}$ and $\mathcal{P}$ which differ only in their (non-Markov) reward distributions. However for each policy π the DPs $\tilde{\mathcal{P}}$ and $\mathcal{P}$ have the same expected return at $t = 0$, that is, $\tilde{v}_0^\pi = v_0^\pi$, or they have the same expected return for every episode.

A2.2.1 Sequence-Markov Decision Processes (SDPs)

We first define decision processes that are Markov except for the reward, which is not required to be Markov.

Definition A3. *A sequence-Markov decision process (SDP) is defined as a finite decision process which is equipped with a Markov policy and has Markov transition probabilities but a reward distribution that is not required to be Markov.*

Proposition A2. *Markov decision processes are sequence-Markov decision processes.*

Proof. MDPs have Markov transition probabilities and are equipped with Markov policies. □

Definition A4. *We call two sequence-Markov decision processes $\mathcal{P}$ and $\tilde{\mathcal{P}}$ that have the same Markov transition probabilities and are equipped with the same Markov policy* sequence-equivalent.

Lemma A2. *Two sequence-Markov decision processes that are sequence-equivalent have the same probability to generate state-action sequences $(s_0, a_0, \ldots, s_t, a_t)$, $0 \leqslant t \leqslant T$.*

Proof. Sequence generation only depends on transition probabilities and policy. Therefore the probability of generating a particular sequences is the same for both SDPs. □

A2.2.2 Return-Equivalent SDPs

We define return-equivalent SDPs which can be shown to have the same optimal policies.

Definition A5. *Two sequence-Markov decision processes $\tilde{\mathcal{P}}$ and $\mathcal{P}$ are* return-equivalent *if they differ only in their reward but for each policy π have the same expected return $\tilde{v}_0^\pi = v_0^\pi$. $\tilde{\mathcal{P}}$ and $\mathcal{P}$ are* strictly return-equivalent *if they have the same expected return for every episode and for each policy π:*

$$\mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \; = \; \mathrm{E}_\pi \left[G_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \; . \tag{A31}$$

The definition of return-equivalence can be generalized to strictly monotonic functions f for which $\tilde{v}_0^\pi = f(v_0^\pi)$. Since strictly monotonic functions do not change the ordering of the returns, maximal returns stay maximal after applying the function f.
Strictly return-equivalent SDPs are return-equivalent as the next proposition states.

Proposition A3. *Strictly return-equivalent sequence-Markov decision processes are return-equivalent.*

Proof. The expected return at $t = 0$ given a policy is the sum of the probability of generating a sequence times the expected reward for this sequence. Both expectations are the same for two strictly return-equivalent sequence-Markov decision processes. Therefore the expected return at time $t = 0$ is the same. □

The next proposition states that return-equivalent SDPs have the same optimal policies.

Proposition A4. *Return-equivalent sequence-Markov decision processes have the same optimal policies.*

Proof. The optimal policy is defined as maximizing the expected return at time $t = 0$. For each policy the expected return at time $t = 0$ is the same for return-equivalent decision processes. Consequently, the optimal policies are the same. □

Two strictly return-equivalent SDPs have the same expected return for each state-action sub-sequence $(s_0, a_0, \ldots, s_t, a_t)$, $0 \leqslant t \leqslant T$.

Lemma A3. *Two strictly return-equivalent SDPs $\tilde{\mathcal{P}}$ and $\mathcal{P}$ have the same expected return for each state-action sub-sequence $(s_0, a_0, \ldots, s_t, a_t)$, $0 \leqslant t \leqslant T$:*

$$\mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_t, a_t \right] \;=\; \mathrm{E}_\pi \left[G_0 \mid s_0, a_0, \ldots, s_t, a_t \right] \;. \tag{A32}$$

Proof. Since the SDPs are strictly return-equivalent, we have

$$\begin{aligned} & \mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_t, a_t \right] \\ &= \sum_{s_{t+1}, a_{t+1}, \ldots, s_T, a_T} p_\pi(s_{t+1}, a_{t+1}, \ldots, s_T, a_T \mid s_t, a_t) \, \mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \\ &= \sum_{s_{t+1}, a_{t+1}, \ldots, s_T, a_T} p_\pi(s_{t+1}, a_{t+1}, \ldots, s_T, a_T \mid s_t, a_t) \, \mathrm{E}_\pi \left[G_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \\ &= \mathrm{E}_\pi \left[G_0 \mid s_0, a_0, \ldots, s_t, a_t \right] \;. \end{aligned} \tag{A33}$$

We used the marginalization of the full probability and the Markov property of the state-action sequence. □

We now give the analog definitions and results for MDPs which are SDPs.

Definition A6. *Two Markov decision processes $\tilde{\mathcal{P}}$ and $\mathcal{P}$ are* return-equivalent *if they differ only in $p(\tilde{r} \mid s, a)$ and $p(r \mid s, a)$ but have the same expected return $\tilde{v}_0^\pi = v_0^\pi$ for each policy π. $\tilde{\mathcal{P}}$ and $\mathcal{P}$ are* strictly return-equivalent *if they have the same expected return for every episode and for each policy π:*

$$\mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \;=\; \mathrm{E}_\pi \left[G_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \;. \tag{A34}$$

Strictly return-equivalent MDPs are return-equivalent as the next proposition states.

Proposition A5. *Strictly return-equivalent decision processes are return-equivalent.*

Proof. Since MDPs are SDPs, the proposition follows from Proposition A3. □

Proposition A6. *Return-equivalent Markov decision processes have the same optimal policies.*

Proof. Since MDPs are SDPs, the proposition follows from Proposition A4. □

For strictly return-equivalent MDPs the expected return is the same if a state-action sub-sequence is given.

Proposition A7. *Strictly return-equivalent MDPs $\tilde{\mathcal{P}}$ and $\mathcal{P}$ have the same expected return for a given state-action sub-sequence $(s_0, a_0, \ldots, s_t, a_t)$, $0 \leqslant t \leqslant T$:*

$$\mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_t, a_t \right] \;=\; \mathrm{E}_\pi \left[G_0 \mid s_0, a_0, \ldots, s_t, a_t \right] \;. \tag{A35}$$

Proof. Since MDPs are SDPs, the proposition follows from Lemma A3. □

A2.3 Reward Redistribution for Strictly Return-Equivalent SDPs

Strictly return-equivalent SDPs $\tilde{\mathcal{P}}$ and $\mathcal{P}$ can be constructed by a reward redistribution.

A2.3.1 Reward Redistribution

We define reward redistributions for SDPs.

Definition A7. *A* reward redistribution *given an SDP $\tilde{\mathcal{P}}$ is a fixed procedure that redistributes for each state-action sequence $s_0, a_0, \ldots, s_T, a_T$ the realization of the associated return variable $\tilde{G}_0 = \sum_{t=0}^T \tilde{R}_{t+1}$ or its expectation $\mathrm{E}\left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_T, a_T \right]$ along the sequence. The redistribution creates a new SDP $\mathcal{P}$ with redistributed reward R_{t+1} at time $(t+1)$ and return variable $G_0 = \sum_{t=0}^T R_{t+1}$. The redistribution procedure ensures for each sequence either $\tilde{G}_0 = G_0$ or*

$$\mathrm{E}_\pi \left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \;=\; \mathrm{E}_\pi \left[G_0 \mid s_0, a_0, \ldots, s_T, a_T \right] \;. \tag{A36}$$

Reward redistributions can be very general. A special case is if the return can be deduced from the past sequence, which makes the return causal.

Definition A8. *A reward redistribution is* causal *if for the redistributed reward R_{t+1} the following holds:*

$$\mathrm{E}\left[R_{t+1} \mid s_0, a_0, \ldots, s_T, a_T\right] \;=\; \mathrm{E}\left[R_{t+1} \mid s_0, a_0, \ldots, s_t, a_t\right] \;. \tag{A37}$$

For our approach we only need reward redistributions that are second order Markov.

Definition A9. *A causal reward redistribution is* second order Markov *if*

$$\mathrm{E}\left[R_{t+1} \mid s_0, a_0, \ldots, s_t, a_t\right] \;=\; \mathrm{E}\left[R_{t+1} \mid s_{t-1}, a_{t-1}, s_t, a_t\right] \;. \tag{A38}$$

A2.4 Reward Redistribution Constructs Strictly Return-Equivalent SDPs

Theorem A1. *If the SDP $\mathcal{P}$ is obtained by reward redistribution from the SDP $\tilde{\mathcal{P}}$, then $\tilde{\mathcal{P}}$ and $\mathcal{P}$ are strictly return-equivalent.*

Proof. For redistributing the reward we have for each state-action sequence $s_0, a_0, \ldots, s_T, a_T$ the same return $\tilde{G}_0 = G_0$, therefore

$$\mathrm{E}_\pi\left[\tilde{G}_0 \mid s_0, a_0, \ldots, s_T, a_T\right] \;=\; \mathrm{E}_\pi\left[G_0 \mid s_0, a_0, \ldots, s_T, a_T\right] \;. \tag{A39}$$

For redistributing the expected return the last equation holds by definition. The last equation is the definition of strictly return-equivalent SDPs. □

The next theorem states that the optimal policies are still the same when redistributing the reward.

Theorem A2. *If the SDP $\mathcal{P}$ is obtained by reward redistribution from the SDP $\tilde{\mathcal{P}}$, then both SDPs have the same optimal policies.*

Proof. According to Theorem A1, the SDP $\mathcal{P}$ is strictly return-equivalent to the SDP $\tilde{\mathcal{P}}$. According to Proposition A3 and Proposition A4 the SDP $\mathcal{P}$ and the SDP $\tilde{\mathcal{P}}$ have the same optimal policies. □

A2.4.1 Special Cases of Strictly Return-Equivalent Decision Processes: Reward Shaping, Look-Ahead Advice, and Look-Back Advice

Redistributing the reward via reward shaping [87, 143], look-ahead advice, and look-back advice [144] is a special case of reward redistribution that leads to MDPs which are strictly return-equivalent to the original MDP. We show that reward shaping is a special case of reward redistributions that lead to MDPs which are strictly return-equivalent to the original MDP. First, we subtract from the potential the constant $c = (\Phi(s_0, a_0) - \gamma^T \Phi(s_T, a_T))/(1 - \gamma^T)$, which is the potential of the initial state minus the discounted potential in the last state divided by a fixed divisor. Consequently, the sum of additional rewards in reward shaping, look-ahead advice, or look-back advice from 1 to T is zero. The original sum of additional rewards is

$$\sum_{i=1}^{T} \gamma^{i-1} \left(\gamma \Phi(s_i, a_i) \;-\; \Phi(s_{i-1}, a_{i-1})\right) \;=\; \gamma^T \Phi(s_T, a_T) \;-\; \Phi(s_0, a_0) \;. \tag{A40}$$

If we assume $\gamma^T \Phi(s_T, a_T) = 0$ and $\Phi(s_0, a_0) = 0$, then reward shaping does not change the return and the shaping reward is a reward redistribution leading to an MDP that is strictly return-equivalent to the original MDP. For $T \to \infty$ only $\Phi(s_0, a_0) = 0$ is required. The assumptions can always be fulfilled by adding a single new initial state and a single new final state to the original MDP.
Without the assumptions $\gamma^T \Phi(s_T, a_T) = 0$ and $\Phi(s_0, a_0) = 0$, we subtract $c = (\Phi(s_0, a_0) - \gamma^T \Phi(s_T, a_T))/(1 - \gamma^T)$ from all potentials Φ, and obtain

$$\sum_{i=1}^{T} \gamma^{i-1} \left(\gamma(\Phi(s_i, a_i) \;-\; c) \;-\; (\Phi(s_{i-1}, a_{i-1}) \;-\; c)\right) \;=\; 0 \;. \tag{A41}$$

Therefore, the potential-based shaping function (the additional reward) added to the original reward does not change the return, which means that the shaping reward is a reward redistribution that leads to an MDP that is strictly return-equivalent to the original MDP. Obviously, reward shaping is a special case of reward redistribution that leads to a strictly return-equivalent MDP. Reward shaping does not change the general learning behavior if a constant c is subtracted from the potential function